ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 23-Sep-2026

Site: Sri. H. R. Vishwanath | Location: Shivamogga, Karnataka | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
8.80 kW	**13,953 kWh**	**Rs. 387,222**	**4Y 6M**
8.80 kWp Installed	CUF 18.1% | PR 75%	Rs. 3.87 Lakhs	Profit: Rs. 41.26L

[SITE] Customer & Site Engineering Parameters

Verified project configuration, contact details, and sizing pathway

Company / Customer Name	Sri. H. R. Vishwanath
Contact Mobile / Phone	+91 94481 23456
Location / Address	Shivamogga, Karnataka
Contact Email Address	vishwanath@voltroof.in
Design Path Selected	Rooftop Area Analysis
Approved Sanctioned Load (CD)	10 kW (KERC / ESCOM Tariff Order)
Rooftop Area Verification	Deficit Rooftop Area (-96 sq.ft deficit, limited to 8.80 kW)
System Target Capacity	8.80 kW (Single LT Installation)
Total Installed Array Rating	8.80 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	16 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	875.00 Sq.Feet (81.29 sq.m)
Utilized Usable Rooftop Area	704 Sq.Feet (65.40 sq.m)
First Year Annual Generation	13,952.76 kWh
25-Year Cumulative Generation	310,000.55 kWh
Estimated Payback Period	4 Years & 6 Months
Post-Payback Profitability (25 Yrs)	Rs. 4,126,239.81 (INR 41.26 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 1,869,392.37 (INR 18.69 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 254 Tonnes CO2 | Tree Equivalence: 406 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Residential Rooftop (Rooftop Array)
Gross Rooftop Area	875.00 Sq.Feet (81.29 sq.m)
Utilized Effective Rooftop Area	704 Sq.Feet (65.40 sq.m)
Building Plant Target Capacity	8.80 kW
Solar PV Module Rating	550 W N-Type TOPCon
Number of Solar Modules	16 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	7 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (7 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	7 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Growatt P3, Fimer On-Grid
Derated AC Design Power (70% Factor)	6.16 kW AC (from 8.80 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx10 Sq.mm
DC Array Isolator Switch	32A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	8.57 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx10 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W N-Type TOPCon	16 Nos.	Rs. 160,000
02	On-grid inverter	7 kW, three-phase (Growatt P3 or equiv.)	1 No.	Rs. 76,222
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 40,000
04	DC solar cable	6 sq.mm copper, PV-rated	100 m	Rs. 8,500
05	AC cable	16 sq.mm copper	55 m	Rs. 13,200
06	MC4 connectors	PV-rated connector pairs	16 pairs	Rs. 4,800
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 10,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 10,500
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 7,500
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 4,500
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 12,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 40,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 387,222.22**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL CAPITAL INVESTMENT REQUIRED

Turnkey Plant BOM Cost: Rs. 387,222 | ESCOM Utility Budget: Rs. 23,012 | Grand Total Outlay: Rs. 410,234 (Rs. 4.10 Lakhs)

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 387,222	**4Y 6M**	**Rs. 41.26L**	**Rs. 18.69L**
12-Item BOM Basis	Break-Even Year	Rs. 4,126,240	Alternative Asset

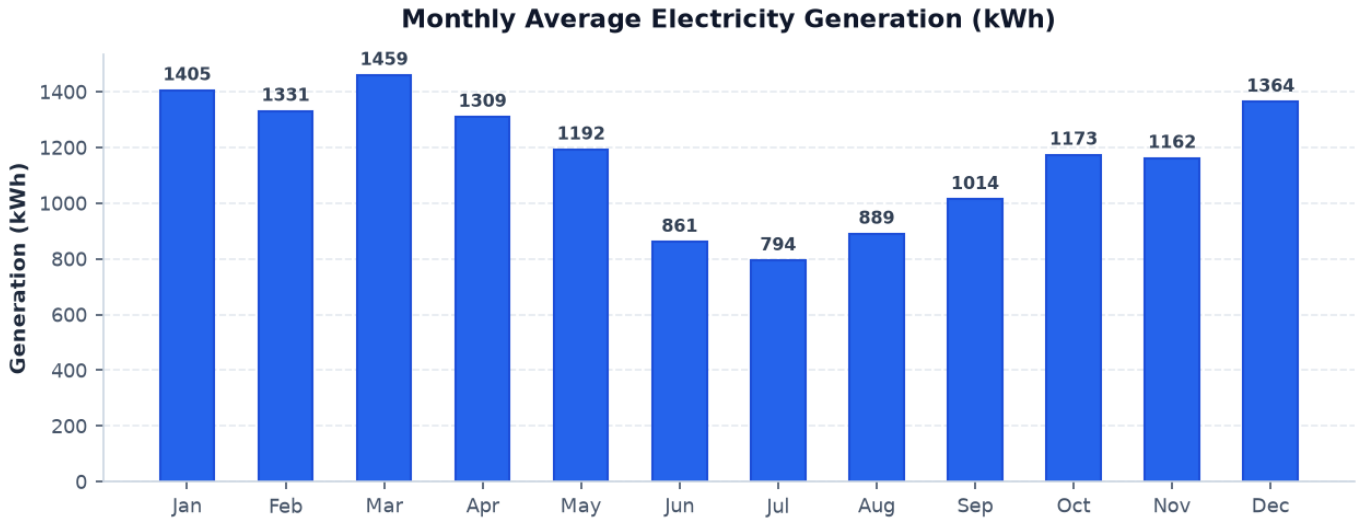
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	13,953	Rs. 8.05	112,320	5,280	107,040	107,040
02	13,813	Rs. 8.45	116,722	5,280	111,442	218,482
03	13,675	Rs. 8.87	121,298	5,280	116,018	334,500
04 *	**13,538**	**Rs. 9.32**	**126,177**	**5,280**	**120,897**	**455,397**
05	13,403	Rs. 9.78	131,081	5,280	125,801	581,198
06	13,269	Rs. 10.27	136,272	5,280	130,992	712,190
07	13,136	Rs. 10.78	141,609	5,280	136,329	848,519
08	13,005	Rs. 11.32	147,215	5,280	141,935	990,454
09	12,875	Rs. 11.89	153,082	5,280	147,802	1,138,256
10	12,746	Rs. 12.48	159,071	5,280	153,791	1,292,047
11	12,619	Rs. 13.11	165,430	5,280	160,150	1,452,197
12	12,492	Rs. 13.76	171,896	5,280	166,616	1,618,813
13	12,368	Rs. 14.45	178,711	5,280	173,431	1,792,244
14	12,244	Rs. 15.18	185,861	5,280	180,581	1,972,825
15	12,121	Rs. 15.93	193,094	5,280	187,814	2,160,639
16	12,000	Rs. 16.74	200,883	5,280	195,603	2,356,242
17	11,880	Rs. 17.57	208,735	5,280	203,455	2,559,697
18	11,761	Rs. 18.45	216,998	5,280	211,718	2,771,415
19	11,644	Rs. 19.37	225,540	5,280	220,260	2,991,675
20	11,527	Rs. 20.34	234,466	5,280	229,186	3,220,861
21	11,412	Rs. 21.36	243,762	5,280	238,482	3,459,342
22	11,298	Rs. 22.43	253,413	5,280	248,133	3,707,475
23	11,185	Rs. 23.55	263,406	5,280	258,126	3,965,601
24	11,073	Rs. 24.73	273,838	5,280	268,558	4,234,159
25	10,962	Rs. 25.96	284,583	5,280	279,303	4,513,462

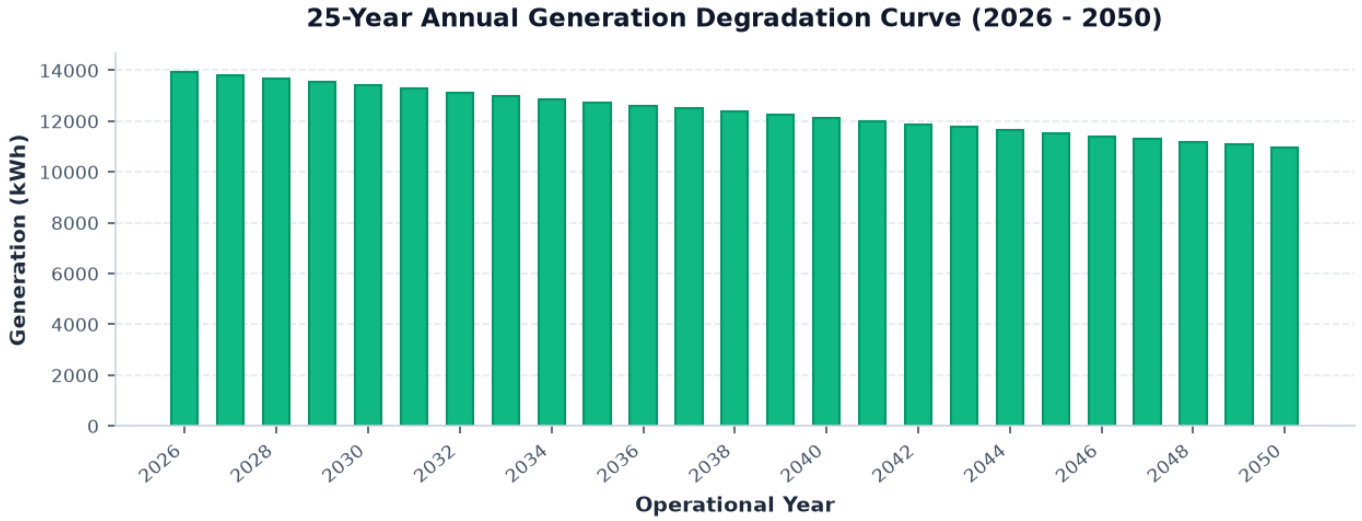
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1586 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

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ON-GRID SOLAR PV SYSTEM

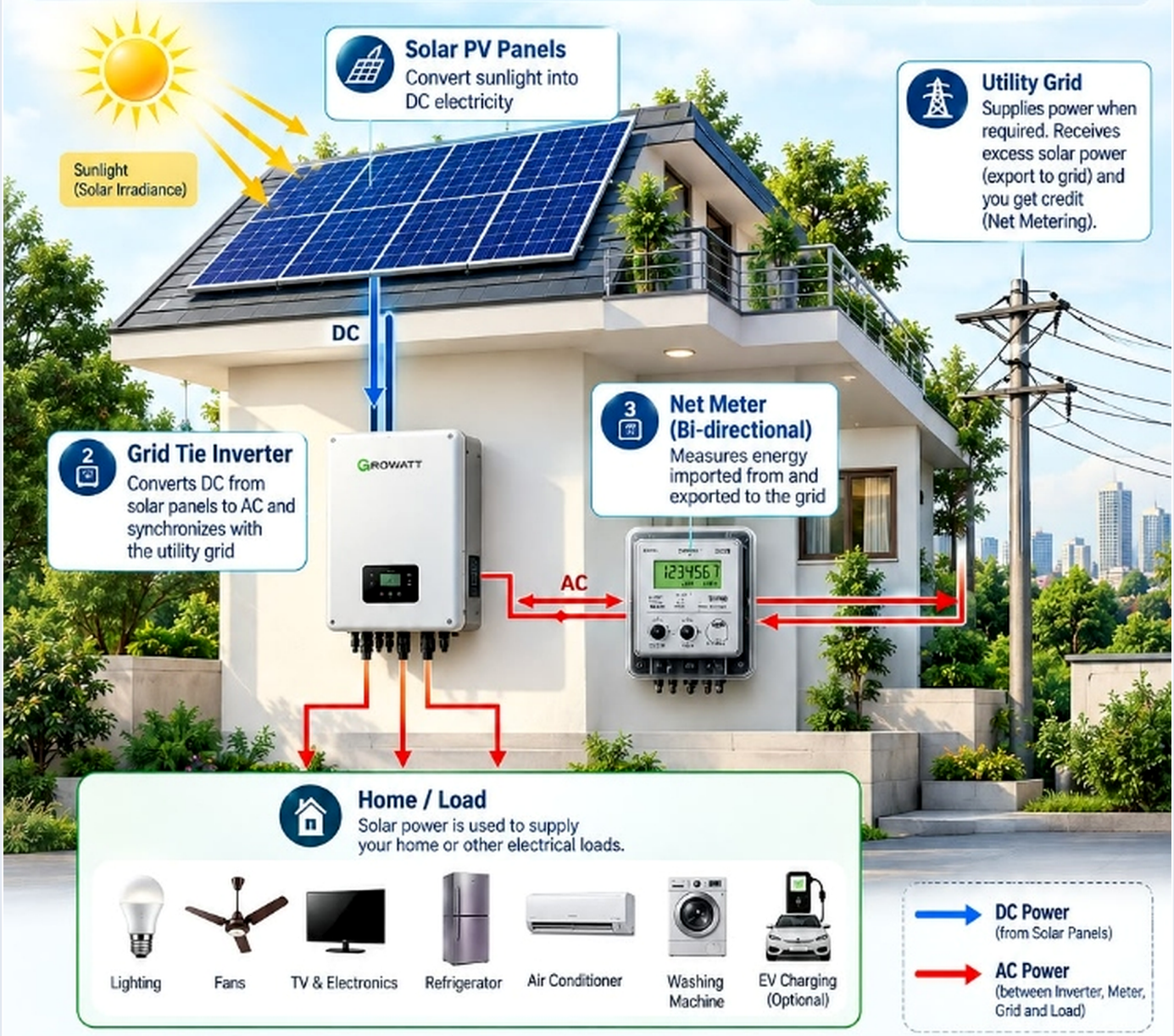
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

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Clean Energy Better Tomorrow